

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

**Duration : 45 Minutes
Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I M. PRASANNA VENKATESH (192573016) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:





Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

ARTIFICIAL INTELLIGENCE

CO-1

AT-3

1) Smart Home Automation System Using Intelligent Agents

A Smart Home Automation System is an Artificial Intelligence (AI) system that automatically controls home appliances such as lights, fans, air conditioners, and security devices. It uses sensors like motion detectors, temperature sensors, light sensors and security cameras to monitor the environment. The system responds to real-time inputs and learns the user's preferences over time to provide better comfort, energy efficiency and security.

1) SIMPLE REFLEX AGENT:

Works only on current input using condition-action rules. Example: turn on lights when motion is detected, turn on AC if temperature exceeds 30°C , activate alarm when smoke is detected. Advantages: Fast, simple, easy to implement. Disadvantages: No memory, cannot learn user behaviour.

2) MODEL-BASED REFLEX AGENT:

Maintains an internal model and remembers previous states. Example: remembers whether a room is occupied and appliance status to avoid unnecessary switching.

3) GOAL - BASED AGENT:

Chooses actions to achieve goals such as comfort, security and energy saving. Example: switches on AC before the user arrives home. Advantages: Intelligent planning and flexibility. Disadvantages: Higher computation and slower decisions.

4) UTILITY - BASED AGENT:

Chooses the action that maximizes overall benefit by balancing comfort, energy usage, security and cost. Example: Increases AC temperature slightly during peak electricity usage to save energy while maintaining comfort. Advantages: Optimal decisions, balances multiple objectives. Disadvantages: Complex to design.

CONCLUSION:

A Hybrid Intelligent Agent provides the best performance because it combines memory, planning and utility optimization, making modern smart homes more comfortable, secure and energy efficient.

4) Analyse the use of Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) for a Robot Navigating an Unknown Maze.

A robot navigating an unknown maze has no prior information about the exit or obstacles. Therefore, it uses uninformed search strategies such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) to explore the maze systematically.

UNIFORM COST SEARCH (UCS):

UCS expands the node with the lowest cumulative path cost first. The robot always chooses the least expensive path from the starting point. It guarantees the optimal solution if path costs are positive. Advantages include optimality and completeness, while disadvantages are high memory usage and increased computation time.

ITERATIVE DEEPENING SEARCH (IDS):

IDS performs repeated depth limited searches by gradually increasing the depth limit until the goal is found. It combines the advantages of DFS and BFS. It uses very little memory and is complete, but repeats searches and does not consider path costs.

COMPARISON:

VCS is best when path costs differ because it finds the least-cost path. IDS is suitable when memory is limited and the goal depth is unknown.

EFFECT OF INTELLIGENT AGENTS:

A Simple Reflex Agent reacts only to current inputs and may revisit the same locations. A Model-Based Agent remembers explored paths and avoids repetition. A Goal-Based Agent plans actions to reach the exit using search algorithms. A Utility-Based Agent evaluates different paths based on time, cost, energy consumption and safety to select the best one.

CONCLUSION:

A Goal-Based Agent combined with Uniform Cost Search is the most suitable solution because it intelligently plans the path and guarantees the least-cost route. IDS is a good alternative when memory is limited.

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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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